

PDE4445

Project Blog and Planning

2025 – 2026

Blog

Motivation

- Practice a different way of presenting your project
- Keep track of progress, changes and developments
- Allows to include multimedia (audio, video, animations)
- Potential worldwide visibility
- Formal following by supervisor and second marker

Content

- Your name
- Project title
- Motivations/main goals
- Background (literature review) including references
- Gantt chart
- Weekly posts on project progress including fundamental steps and insights
- Final outcome video (or demo, if possible)

Suggestions

- Be clear
- Use a mix of text and media – Take advantage of multiple media types
- Document your progress regularly
- You can include the blog in your more complete personal webpage (or GitHub)

Planning

First Steps

- Start your project blog (**Deadline soon!**)
- Start researching the subject of your project:
 - What have people done in similar works?
 - What resources are needed?
 - What should I study?

Planning – First Steps

- **Goal** – what you want to achieve

“A social robot with expressions and verbal communication”

- **Objective** – what your system will look like or be able to do

“Robot head with some motors for showing expressions, microphone and speakers, and a system for generating speech and showing appropriate expressions”

- **Task** – what you need to do to achieve the objectives

“Implement speech recognition and generation, expression patterns; assemble components into a sensible structure, with motors for expressions; design, ordering, assembling, coding, testing, etc.”

Gantt Chart

- Time-based representation of project plan
- Used to make and update a plan
- Made of tasks and milestones (measurable achievements)
- Can specify task and subtasks, timing and dependencies

Gantt Chart

- It is for you – help planning and manage time during project development
- It is about the future – keep it updated – it will change!
- Time estimates; realism; parallel activities
- Do a first draft in any case, improve it later

Task Dependencies

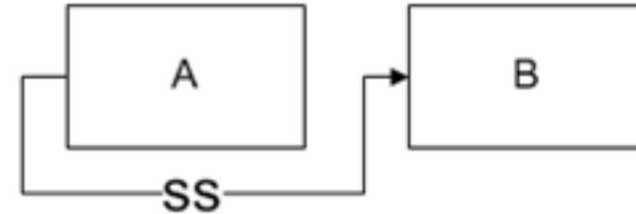
Finish to Start

Task B can't start until Task A has finished



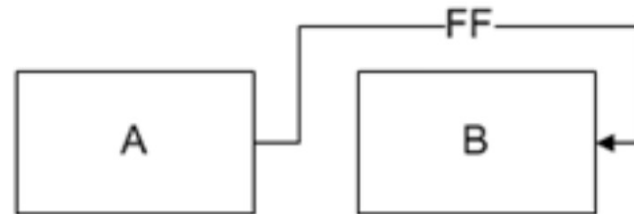
Start to Start

Task B can't start until Task A has started



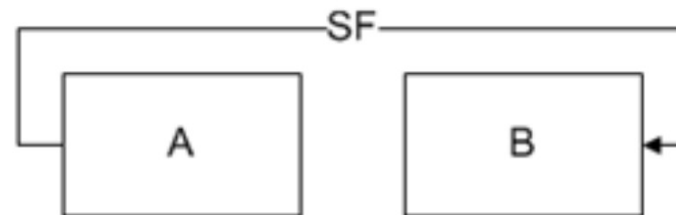
Finish to Finish

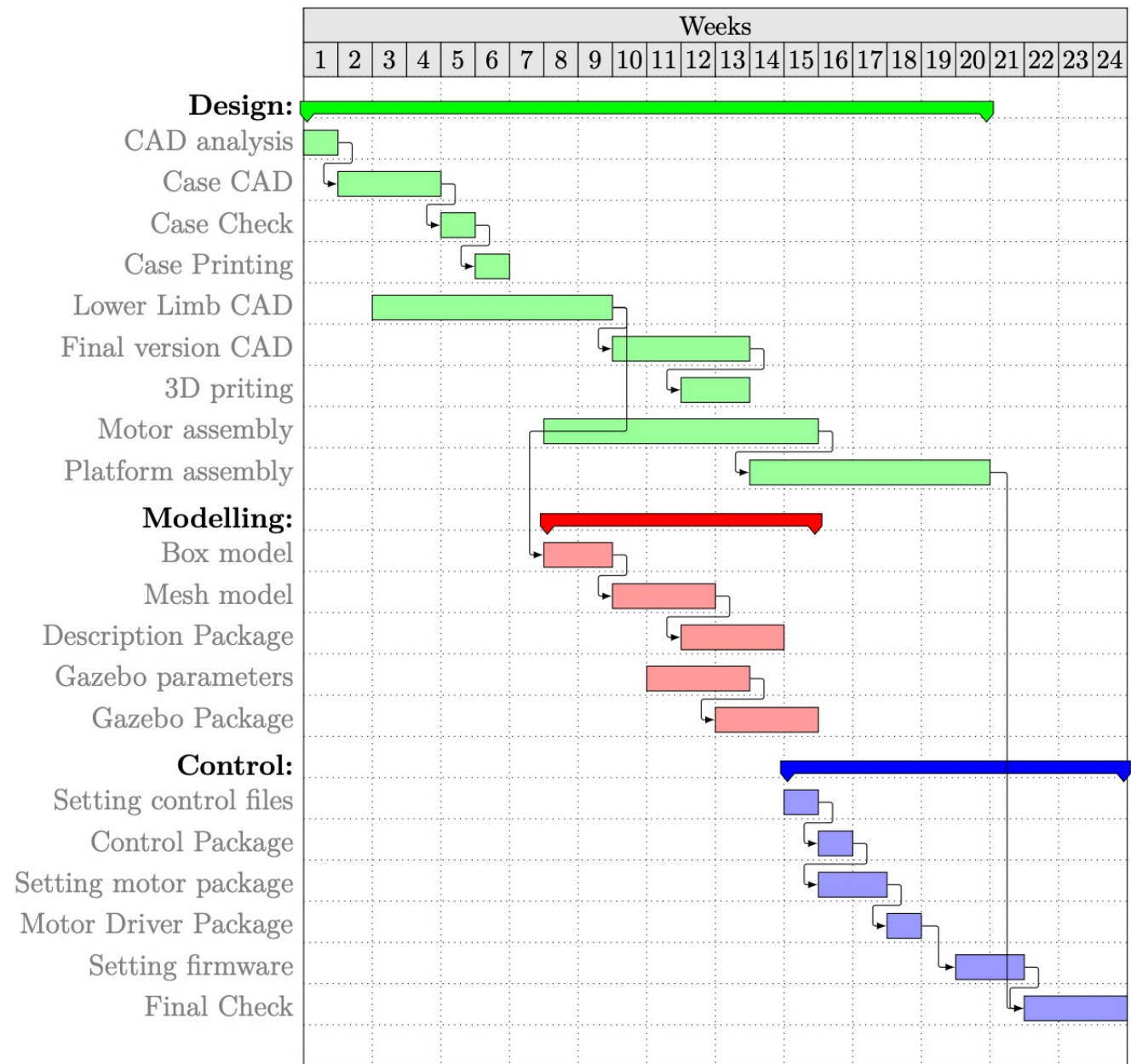
Task B can't finish until Task A has finished



Start to Finish

Task B can't finish until Task A has Started





Planning

1. Establish the project's main goals and objectives

- What do you aim to achieve? What do you expect your final system to be able to do?

2. Split your objectives into smaller ones

- Partial objectives should be increasingly complex and ambitious. Include easy, likely and ambitious outcomes.

3. Translate objectives into tasks

- What do you need to do to achieve the above objectives? They can each require more than one task.

Planning

4. Divide tasks into subtasks

- Find subtasks: The smaller work components that make a task. Work backwards: What is needed before the next subtask can begin?

5. Time your plan

- How long does each subtask take? Do you have enough time?

6. Review and update your plan

- Each of the above steps affect previous ones. Make your plan coherent.

Planning – Phase 1: Project Goals

Establish the main goal(s) of the project.

Talk to your supervisor, and agree on what the fundamental goals are

Planning – Phase 2: Objectives

Start from the project goal(s).

Define the main objectives of your project. They need to be:

Specific, precise.

Observable or measurable.

Interesting, but also feasible.

Planning – Phase 3: Task Creation

1. Review the project objectives you wrote, making sure it's what you want.
2. Define all the tasks you think are required to achieve every objective in the list.
3. Split tasks into lower-level, practical sub-tasks and components.
4. If you cannot make a reasonable guess about how long it will take you to perform a task, you probably need to split it further.

Planning – Phase 4: Gantt Chart

1. Review the list of tasks
2. Estimate the time you need for developing each task.
3. Think about how you can objectively check that a task has actually been completed, and the corresponding objective achieved. This is a milestone.
4. Prepare the Gantt chart of your project, including tasks and milestones, and dependencies if necessary.

First Blog Update

Include your goals and objectives.

You can put them as an abstract, or as first post, but making it always available on top.